

Math 104 — Homework 10

Noam Michael

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Rudin Chapter 7

Problem 1 (7.6). *Prove that the series*

$$\sum_{n=1}^{\infty} (-1)^n \frac{x^2 + n}{n^2}$$

converges uniformly in every bounded interval, but does not converge absolutely for any value of x .

Proof. Split the n th term:

$$(-1)^n \frac{x^2 + n}{n^2} = (-1)^n \frac{x^2}{n^2} + (-1)^n \frac{1}{n}.$$

Uniform convergence on $[-M, M]$. For the first piece, on any bounded interval $|x| \leq M$,

$$\left| (-1)^n \frac{x^2}{n^2} \right| \leq \frac{M^2}{n^2},$$

and $\sum M^2/n^2 < \infty$, so by the Weierstrass M -test, $\sum (-1)^n x^2/n^2$ converges uniformly on $[-M, M]$.

The second piece $\sum (-1)^n/n$ is the alternating harmonic series, a convergent constant (no x -dependence), so its partial sums trivially converge uniformly.

The sum of two uniformly convergent series is uniformly convergent, so $\sum (-1)^n \frac{x^2 + n}{n^2}$ converges uniformly on $[-M, M]$.

No absolute convergence. For any fixed x ,

$$\left| (-1)^n \frac{x^2 + n}{n^2} \right| = \frac{x^2 + n}{n^2} = \frac{x^2}{n^2} + \frac{1}{n}.$$

The series $\sum x^2/n^2$ converges, but $\sum 1/n = \infty$. Hence

$$\sum_{n=1}^{\infty} \left| (-1)^n \frac{x^2 + n}{n^2} \right| = \infty$$

for every x , so the series does not converge absolutely for any x . □

Problem 2 (7.7). *For $n = 1, 2, 3, \dots$, x real, put*

$$f_n(x) = \frac{x}{1 + nx^2}.$$

Show that $\{f_n\}$ converges uniformly to a function f , and that the equation

$$f'(x) = \lim_{n \rightarrow \infty} f'_n(x)$$

is correct if $x \neq 0$, but false if $x = 0$.

Proof. Uniform convergence to $f \equiv 0$. By the AM–GM inequality, for $x \neq 0$,

$$\frac{1}{x} + nx \geq 2\sqrt{n},$$

so $|f_n(x)| = \frac{|x|}{1+nx^2} = \frac{1}{1/|x|+n|x|} \leq \frac{1}{2\sqrt{n}}$. At $x = 0$, $f_n(0) = 0$. Equality holds when $x = \pm 1/\sqrt{n}$, so

$$\|f_n\|_\infty = \frac{1}{2\sqrt{n}} \rightarrow 0.$$

Hence $f_n \rightarrow 0$ uniformly, and $f \equiv 0$.

Derivative interchange. A direct computation gives

$$f'_n(x) = \frac{1 - nx^2}{(1 + nx^2)^2}.$$

At $x = 0$: $f'_n(0) = \frac{1}{1} = 1$ for every n , so $\lim_{n \rightarrow \infty} f'_n(0) = 1$. But $f'(0) = 0$, so $f'(0) \neq \lim f'_n(0)$.

At $x \neq 0$: For large n , the numerator is dominated by $-nx^2$ and the denominator by n^2x^4 , giving

$$f'_n(x) = \frac{1 - nx^2}{(1 + nx^2)^2} \sim \frac{-nx^2}{n^2x^4} = \frac{-1}{nx^2} \rightarrow 0.$$

Since $f'(x) = 0$, we have $f'(x) = \lim_{n \rightarrow \infty} f'_n(x)$ for all $x \neq 0$. □

Problem 3 (7.8). If

$$I(x) = \begin{cases} 0 & (x \leq 0), \\ 1 & (x > 0), \end{cases}$$

if $\{x_n\}$ is a sequence of distinct points of (a, b) , and if $\sum |c_n|$ converges, prove that the series

$$f(x) = \sum_{n=1}^{\infty} c_n I(x - x_n) \quad (a \leq x \leq b)$$

converges uniformly, and that f is continuous for every $x \neq x_n$.

Proof. Uniform convergence. For every $x \in [a, b]$ and every n , $I(x - x_n) \in \{0, 1\}$, so

$$|c_n I(x - x_n)| \leq |c_n|.$$

Since $\sum |c_n| < \infty$, the Weierstrass M -test (with $M_n = |c_n|$) gives uniform convergence of $\sum c_n I(x - x_n)$ on $[a, b]$.

Continuity at $x \neq x_n$. Fix $x_0 \neq x_n$ for all n . For each n , either $x_0 > x_n$ or $x_0 < x_n$. In either case, $I(x - x_n)$ is constant in a neighborhood of x_0 (equal to 1 or 0, respectively), so $c_n I(x - x_n)$ is continuous at x_0 .

Since each term is continuous at x_0 and the series converges uniformly, Theorem 7.12 implies f is continuous at x_0 . □

Problem 4 (7.10). Letting (x) denote the fractional part of the real number x , consider the function

$$f(x) = \sum_{n=1}^{\infty} \frac{(nx)}{n^2} \quad (x \text{ real}).$$

Find all discontinuities of f , and show that they form a countable dense set. Show that f is nevertheless Riemann-integrable on every bounded interval.

Proof. Uniform convergence. Since $0 \leq (t) < 1$ for all t , we have $|(nx)/n^2| \leq 1/n^2$. Since $\sum 1/n^2 < \infty$, the Weierstrass M -test gives uniform convergence of the series on all of $\mathbb{R}$.

Discontinuities at rationals. The fractional part function (t) is discontinuous precisely at integer values of t . So the n th term $(nx)/n^2$ is discontinuous at $x = k/n$ for each integer k , i.e. at every rational with denominator dividing n . At a rational $x = p/q$ (in lowest terms), the term with $n = q$ has a jump discontinuity of size $1/q^2$, and this jump survives in the sum f since the remaining terms are either continuous at p/q or have jumps of the same sign (each (t) jumps downward by 1 at integers). Hence f is discontinuous at every rational.

Continuity at irrationals. If x is irrational, then nx is never an integer for any $n \geq 1$, so every term $(nx)/n^2$ is continuous at x . Since each term is continuous at x and the series converges uniformly, Theorem 7.12 implies f is continuous at x .

Countable dense set. The set of discontinuities is exactly $\mathbb{Q}$, which is countable and dense in $\mathbb{R}$.

Riemann integrability. Each partial sum $S_N(x) = \sum_{n=1}^N (nx)/n^2$ is a finite sum of bounded functions, each having only finitely many discontinuities in any bounded interval, so $S_N \in R$ on every bounded interval. Since $S_N \rightarrow f$ uniformly, Theorem 7.16 implies $f \in R$ on every bounded interval. $\square$

Problem 5 (7.13). Assume that $\{f_n\}$ is a sequence of monotonically increasing functions on $\mathbb{R}$ with $0 \leq f_n(x) \leq 1$ for all x and all n .

(a) Prove that there is a function f and a sequence $\{n_k\}$ such that

$$f(x) = \lim_{k \rightarrow \infty} f_{n_k}(x)$$

for every $x \in \mathbb{R}$. (Helly's selection theorem.)

(b) If, moreover, f is continuous, prove that $f_{n_k} \rightarrow f$ uniformly on compact sets.

Proof. (a) Enumerate the rationals as $r_1, r_2, r_3, \dots$. Since $\{f_n(r_1)\}$ is a sequence in $[0, 1]$, the Bolzano–Weierstrass theorem gives a subsequence $\{f_{n_k^{(1)}}\}$ converging at r_1 . From this, extract a further subsequence $\{f_{n_k^{(2)}}\}$ converging at r_2 (it still converges at r_1). Continuing, we obtain nested subsequences $\{f_{n_k^{(j)}}\}$, the j th converging at $r_1, \dots, r_j$.

Take the diagonal subsequence $g_k := f_{n_k^{(k)}}$. For each j , $\{g_k\}_{k \geq j}$ is a subsequence of $\{f_{n_k^{(j)}}\}$, so $g_k(r_j) \rightarrow g(r_j)$ for some value $g(r_j)$. Thus $\{g_k\}$ converges at every rational.

Now define, for all $x \in \mathbb{R}$,

$$f(x) = \sup\{g(r) : r \in \mathbb{Q}, r \leq x\}.$$

Since each f_n is increasing and $0 \leq f_n \leq 1$, the limit g is increasing and bounded on $\mathbb{Q}$, so f is well-defined, increasing, and takes values in $[0, 1]$.

Convergence at continuity points of f . Fix x where f is continuous. Given $\varepsilon > 0$, choose rationals $r < x < s$ with $f(s) - f(r) < \varepsilon$. Since each g_k is increasing,

$$g_k(r) \leq g_k(x) \leq g_k(s).$$

As $k \rightarrow \infty$, $g_k(r) \rightarrow g(r) = f(r)$ and $g_k(s) \rightarrow g(s) = f(s)$. Also $f(r) \leq f(x) \leq f(s)$, so

$$|g_k(x) - f(x)| \leq f(s) - f(r) + \varepsilon_k < \varepsilon + \varepsilon_k$$

for large k , where $\varepsilon_k \rightarrow 0$. Hence $g_k(x) \rightarrow f(x)$.

Convergence at discontinuity points. Since f is monotone, it has at most countably many discontinuities $\{d_1, d_2, \dots\}$. Apply Bolzano–Weierstrass again: from $\{g_k\}$, extract a subsequence converging at d_1 , then a further subsequence converging at d_2 , and so on. A final diagonal argument yields a subsequence $\{f_{n_k}\}$ converging at every d_j and (being a subsequence of $\{g_k\}$) at every continuity point of f as well. Hence $f_{n_k}(x) \rightarrow f(x)$ for every $x \in \mathbb{R}$.

(b) Suppose f is continuous. Then the argument above applies at every point (there are no discontinuities to handle separately), and $g_k \rightarrow f$ pointwise on $\mathbb{R}$. Fix a compact set $[a, b]$. Given $\varepsilon > 0$, by uniform continuity of f on $[a, b]$, choose rationals $a = r_0 < r_1 < \dots < r_m = b$ with $f(r_i) - f(r_{i-1}) < \varepsilon$ for each i . For $x \in [r_{i-1}, r_i]$, monotonicity gives

$$g_k(r_{i-1}) \leq g_k(x) \leq g_k(r_i),$$

and $f(r_{i-1}) \leq f(x) \leq f(r_i)$. Since $g_k(r_j) \rightarrow f(r_j)$ for each of the finitely many r_j , there exists K such that $|g_k(r_j) - f(r_j)| < \varepsilon$ for all $k \geq K$ and all j . Then for $x \in [r_{i-1}, r_i]$,

$$|g_k(x) - f(x)| \leq f(r_i) - f(r_{i-1}) + 2\varepsilon < 3\varepsilon$$

for all $k \geq K$, uniformly in $x \in [a, b]$. □

Problem 6 (7.14). Let f be a continuous real function on $\mathbb{R}$ with the following properties: $0 \leq f(t) \leq 1$, $f(t+2) = f(t)$ for every t , and

$$f(t) = \begin{cases} 0 & (0 \leq t \leq \frac{1}{3}), \\ 1 & (\frac{2}{3} \leq t \leq 1). \end{cases}$$

Put $\Phi(t) = (x(t), y(t))$, where

$$x(t) = \sum_{n=1}^{\infty} 2^{-n} f(3^{2n-1}t), \quad y(t) = \sum_{n=1}^{\infty} 2^{-n} f(3^{2n}t).$$

Prove that Φ is continuous and that Φ maps $I = [0, 1]$ onto the unit square $I^2 \subset \mathbb{R}^2$. In fact, show that Φ maps the Cantor set onto I^2 .

Proof. Continuity. Each term $2^{-n}f(3^kt)$ is continuous (composition of continuous functions, scaled by 2^{-n}), and $|2^{-n}f(3^kt)| \leq 2^{-n}$ since $0 \leq f \leq 1$. Since $\sum 2^{-n} < \infty$, the Weierstrass M -test gives uniform convergence of both $x(t)$ and $y(t)$. By Theorem 7.12, $x(t)$ and $y(t)$ are continuous, hence Φ is continuous.

Surjectivity (onto I^2 via the Cantor set). Let $(x_0, y_0) \in [0, 1]^2$. Write binary expansions

$$x_0 = \sum_{n=1}^{\infty} 2^{-n} a_{2n-1}, \quad y_0 = \sum_{n=1}^{\infty} 2^{-n} a_{2n},$$

where each $a_i \in \{0, 1\}$. Define

$$t_0 = \sum_{i=1}^{\infty} \frac{2a_i}{3^{i+1}}.$$

Since each digit $2a_i \in \{0, 2\}$, t_0 has only digits 0 and 2 in its base-3 expansion, so t_0 belongs to the Cantor set.

We claim $f(3^k t_0) = a_k$ for every $k \geq 1$. Indeed,

$$3^k t_0 = \sum_{i=1}^{\infty} \frac{2a_i}{3^{i+1-k}} = \sum_{i=1}^k 2a_i \cdot 3^{k-i-1} + \frac{2a_{k+1}}{3^2} + \frac{2a_{k+2}}{3^3} + \dots$$

The first sum is an even integer (call it $2m$), and the remaining tail lies in $[0, 2/9 + 2/27 + \dots] = [0, 1/3]$. So the fractional part of $3^k t_0$ is $\frac{2a_{k+1}}{3^2} + \dots$; but we need f evaluated at $3^k t_0$.

More precisely, $3^k t_0 = 2m + \frac{2a_k}{3} + r$ where $0 \leq r \leq 1/3$. Since f has period 2, $f(3^k t_0) = f(\frac{2a_k}{3} + r)$. If $a_k = 0$, then $\frac{2a_k}{3} + r \in [0, 1/3]$, so $f = 0 = a_k$. If $a_k = 1$, then $\frac{2a_k}{3} + r \in [2/3, 1]$, so $f = 1 = a_k$.

Therefore

$$\begin{aligned} x(t_0) &= \sum_{n=1}^{\infty} 2^{-n} f(3^{2n-1} t_0) = \sum_{n=1}^{\infty} 2^{-n} a_{2n-1} = x_0, \\ y(t_0) &= \sum_{n=1}^{\infty} 2^{-n} f(3^{2n} t_0) = \sum_{n=1}^{\infty} 2^{-n} a_{2n} = y_0. \end{aligned}$$

So $\Phi(t_0) = (x_0, y_0)$, and since t_0 is in the Cantor set, Φ maps the Cantor set onto I^2 . $\square$

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